

Music Box Doll Builder — Phase 1 Technical Specification

Summary

The app is a virtual music-box builder for iOS, targeting users who appreciate aesthetic and artistic content. Users assemble a custom doll from a curated catalog of 3D-scanned parts created by a specialist doll artist, choosing elements slot by slot (head, hair, accessories, body, limbs, base) with constrained transforms to preserve visual quality. Once assembled, the doll is placed inside a stylized music-box scene with a background, foreground decorations, and a music track, then rendered as a short looping video for sharing via the native iOS share sheet. Monetization begins after the user's first successful export/share, with a paywall gating subsequent use. This spec covers the Phase 1 prototype: one doll template, approximately 20–30 assets across 3–5 slots, one scene, one music track, local on-device rendering, and a simple paywall trigger.

Requirements

- **REQ-01** — The app shall launch with an animated landing screen showing a pre-built doll that invites the user into the creation flow.
- **REQ-02** — The app shall present a step-by-step slot-based doll builder, progressing through head, hair, hat, horns/halo, body shell, inner insert, collar, wings, hands, sleeves, lower body, feet/base, and tail — with each slot showing a scrollable catalog of available assets.
- **REQ-03** — Every slot shall include an empty/none option, allowing the user to skip any category.
- **REQ-04** — Users shall be able to navigate backward and forward between slots at any time during the build flow.
- **REQ-05** — Each asset shall snap to a predefined anchor point and support constrained rotation (X, Y, and around own axis) and constrained scale (within min/max bounds defined per slot) — no freeform positioning for core body parts.
- **REQ-06** — After completing the slot-by-slot editor, the app shall present a global adjustment mode where the user can tweak scale, rotation, and position of placed elements within enforced bounds.
- **REQ-07** — After doll editing, the app shall present catalogs for background, foreground, and floating decorative props to style the music-box scene.
- **REQ-08** — After scene decoration, the app shall present a music selection screen with a catalog of music-box-style tracks.

- **REQ-09** — The app shall render a short looping video on-device at 1080×1920 (9:16 vertical) showing the doll rotating inside the music box with the selected music track.
- **REQ-10** — The app shall open the native iOS share sheet to let the user share/export the rendered video to Instagram Reels, Stories, or any supported destination.
- **REQ-11** — The first export/share shall be free; after the first successful export, a paywall shall block further exports until the user completes a one-time in-app purchase via StoreKit 2.
- **REQ-12** — The app shall allow saving draft projects locally (maximum 5) and resuming them in a later session.
- **REQ-13** — The Phase 1 app shall ship with one doll template, 3-5 active slots, 20-30 bundled assets, one scene template, and one music track — all included in the app bundle with no network required to build and export.
- **REQ-14** — The entire Phase 1 flow shall work without user account creation or sign-in (guest-only).
- **REQ-15** — The app shall support restoring the one-time purchase via StoreKit 2 restore functionality.

Technical Stack

Layer	Technology / Package	Notes
iOS UI	Swift, SwiftUI	iOS 17+ minimum deployment target
3D Engine	RealityKit	Scene composition, slot anchoring, constrained transforms, on-device render
3D Assets	USDZ	Native RealityKit format; convert from scans via Reality Converter or Blender
Video Export	AVFoundation	Composites rendered frames + audio into 1080×1920 H.264/AAC .mp4
Monetization	StoreKit 2	One-time purchase, local receipt validation, restore support
Persistence	SwiftData	Local draft project save/load (iOS 17+ native)
Sharing	UIActivityViewController	Native iOS share sheet for Reels/Stories/any target
Analytics	TelemetryDeck	Privacy-friendly, GDPR-compliant, lightweight Swift SDK, free tier

Layer	Technology / Package	Notes
CI/CD	Xcode Cloud	25 free hours/month with Apple Developer Program; TestFlight distribution
Backend	None (Phase 1)	Deferred to Phase 2; all data and assets on-device
Asset Pipeline	Blender + Reality Converter	Scan cleanup, retopo, UV/texture optimization, USDZ export, anchor/pivot setup

Architecture

```
mermaid
graph TD
    A[App Entry / Landing] --> B[Editor Module]
    B --> C[Scene Composer Module]
    C --> D[Music Selector]
    D --> E[Render Engine Module]
    E --> F[Paywall Module]
    F -->|paid or first free| G[Share Flow]

    B --> H[Catalog Module]
    C --> H
    D --> H

    B --> I[Project Store Module]
    C --> I
    D --> I

    F --> J[StoreKit 2]
    E --> K[AVFoundation]
    B --> L[RealityKit Scene]
    C --> L
    E --> L

    M[Analytics Module] -.-> B
    M -.-> C
    M -.-> E
    M -.-> F
    M -.-> G
```

Components

App Entry / Landing

- **Responsibility:** Animated landing screen with pre-built doll; entry point into creation flow or draft resume
- **Exposes:** Navigation to Editor or saved draft list
- **Depends on:** Project Store (to check for existing drafts), Catalog (for landing doll assets), RealityKit Scene

Editor Module

- **Responsibility:** Step-by-step slot-based doll builder; manages slot navigation (forward/back), asset selection per slot, constrained transform controls (rotate X/Y/axis, scale within bounds), and global adjustment mode
- **Exposes:** Current doll composition state (selected assets + transforms per slot), slot navigation API
- **Depends on:** Catalog Module (asset lists + metadata per slot), Project Store (autosave), RealityKit Scene (3D preview)

Scene Composer Module

- **Responsibility:** Background, foreground, and floating prop selection and placement after doll editing is complete
- **Exposes:** Scene decoration state (selected background, foreground, props + transforms)
- **Depends on:** Catalog Module, Project Store, RealityKit Scene

Music Selector

- **Responsibility:** Presents music track catalog, plays preview, stores selection
- **Exposes:** Selected track reference and metadata (duration, file reference)
- **Depends on:** Catalog Module, Project Store

Catalog Module

- **Responsibility:** Loads and provides asset metadata and USDZ references for all slots, scenes, props, and music; flat list per slot with no subcategories in Phase 1
- **Exposes:** Typed asset manifests per slot category, including asset ID, display name, preview thumbnail, USDZ file reference, default transform, min/max scale, min/max rotation, anchor point, dependencies, exclusions
- **Depends on:** Bundled asset files in app package

Project Store Module

- **Responsibility:** Persists and restores draft projects using SwiftData; captures full composition state (all slot selections, transforms, scene choices, music choice, current editor step)
- **Exposes:** Save, load, list, delete draft operations
- **Depends on:** SwiftData (iOS 17+)

Render Engine Module

- **Responsibility:** Captures the composed RealityKit scene as a 360° rotation video at 1080×1920; composites audio track; outputs .mp4; user-configurable duration (default 10s)
- **Exposes:** Render trigger, progress callback, output file URL
- **Depends on:** RealityKit Scene (frame capture), AVFoundation (video + audio composition)

Paywall Module

- **Responsibility:** Tracks whether the user has used their free export; presents paywall on second+ export; handles one-time purchase and restore via StoreKit 2
- **Exposes:** Entitlement check (canExport → bool), purchase flow trigger, restore flow trigger
- **Depends on:** StoreKit 2, UserDefaults (free export flag), Transaction.currentEntitlements (restore)

Share Flow

- **Responsibility:** Opens native iOS share sheet with rendered video file
- **Exposes:** Share trigger accepting video file URL
- **Depends on:** UIActivityViewController

Analytics Module

- **Responsibility:** Lightweight event tracking across the funnel; privacy-friendly, no PII
- **Exposes:** Track event API (event name + optional properties)
- **Depends on:** TelemetryDeck SDK

Patterns

- **Modular Swift Packages** — Each component is a distinct Swift package or target within the Xcode project, enabling clean dependency boundaries and testability without over-engineering into microservices

- **Slot-based composition model** — The doll is defined as a typed array of slots, each holding an optional asset reference and transform state; this makes save/load, validation, and rendering deterministic
- **Constrained transform envelope** — Every asset carries metadata defining allowed transform ranges; the Editor enforces these bounds at the UI layer, preventing broken compositions
- **Entitlement-gated export** — The Paywall module wraps the render-to-share pipeline with a single `canExport` check, keeping monetization logic isolated from the creative flow
- **Offline-first** — All assets bundled; all persistence local via SwiftData; no network dependency in Phase 1

Data Model

DollProject (SwiftData)

Field	Type	Description
id	UUID	Unique project identifier
name	String?	Optional user label
createdAt	Date	Creation timestamp
updatedAt	Date	Last modified timestamp
currentStep	SlotType?	Editor step to resume from
slotSelections	[SlotSelection]	Array of slot → asset + transform mappings
sceneBackground	AssetReference?	Selected background
sceneForeground	AssetReference?	Selected foreground
sceneProps	[PropPlacement]	Placed floating decorations
musicTrackID	String?	Selected music track reference
videoDuration	Double	Render duration in seconds (default 10.0)

SlotSelection

Field	Type	Description
slotType	SlotType	Enum: head, hair, hat, horns, halo, bodyShell, innerInsert, collar, wings, leftHand, rightHand, leftSleeve, rightSleeve, lowerBody, feetBase, tail
assetID	String?	nil = skipped slot
position	SIMD3<Float>	Offset from anchor
rotation	SIMD3<Float>	Euler rotation (bounded)
scale	Float	Uniform scale (bounded)

AssetManifestEntry (read-only, bundled JSON)

Field	Type	Description
assetID	String	Unique asset identifier
slotType	SlotType	Which slot this belongs to
displayName	String	Label shown in catalog
previewImage	String	Thumbnail filename
usdzFile	String	USDZ filename in bundle
defaultTransform	Transform	Default position/rotation/scale
minScale	Float	Lower scale bound
maxScale	Float	Upper scale bound
minRotation	SIMD3<Float>	Lower rotation bounds per axis
maxRotation	SIMD3<Float>	Upper rotation bounds per axis
anchorPoint	SIMD3<Float>	Snap-to point in scene
excludes	[String]	Asset IDs incompatible with this one
dependencies	[String]	Asset IDs required alongside this one

Flows

1. App Launch Flow

mermaid

```
sequenceDiagram
    participant U as User
    participant L as Landing Screen
    participant PS as Project Store

    U->>L: Opens app
    L->>PS: Check for saved drafts
    PS-->>L: Draft list (may be empty)
    L->>U: Play landing animation (pre-built doll)
    L->>U: Show "Create New" button
    alt Has saved drafts
        L->>U: Show "Resume Draft" option
        U->>L: Taps "Resume Draft"
        L->>PS: Load selected draft
        PS-->>L: DollProject state
        L->>U: Navigate to Editor at saved step
    else New creation
        U->>L: Taps "Create New"
        L->>U: Navigate to Editor (step 1)
    end
end
```

Steps:

1. **User → Landing Screen** — Opens the app; landing animation plays showing a pre-built doll
2. **Landing Screen → Project Store** — Checks for existing saved drafts
3. **Landing Screen → User** — Displays "Create New" button and, if drafts exist, a "Resume Draft" option
4. **User → Landing Screen** — Taps either "Create New" (starts fresh at first slot) or "Resume Draft" (loads saved state and jumps to the saved editor step)

2. Slot-by-Slot Editor Flow

mermaid

sequenceDiagram

```
participant U as User
participant E as Editor Module
participant C as Catalog Module
participant R as RealityKit Scene
participant PS as Project Store

E->>C: Request assets for current slot
C-->>E: Asset list + metadata
E->>U: Show slot catalog (scrollable) + 3D preview
U->>E: Selects asset (or "none")
E->>R: Place asset at anchor with default transform
R-->>U: Updated 3D preview
U->>E: Adjusts rotation/scale (within bounds)
E->>R: Update transform (clamped)
E->>PS: Autosave current state
alt Taps "Next"
    E->>E: Advance to next slot
else Taps "Back"
    E->>E: Return to previous slot
end
Note over E: Repeats for all active slots
E->>U: All slots complete → show full doll with "Continue" and "Back to slots" c
```

Steps:

1. **Editor → Catalog** — Requests available assets for the current slot type
 2. **Editor → User** — Displays scrollable catalog sidebar with thumbnails; each catalog includes a "none" option to skip
 3. **User → Editor** — Selects an asset or "none"
 4. **Editor → RealityKit Scene** — Places selected asset at predefined anchor with default transform; updates 3D preview
 5. **User → Editor** — Optionally adjusts rotation (X, Y, own axis) and scale; Editor clamps values to metadata bounds
 6. **Editor → Project Store** — Autosaves composition state after each change
 7. **User → Editor** — Taps "Next" to advance to next slot, or "Back" to revisit a previous slot
 8. **Editor → User** — After final slot, shows full assembled doll with option to continue to global adjustment or go back to any slot
-

3. Global Adjustment Flow

mermaid

```
sequenceDiagram
    participant U as User
    participant GA as Global Adjustment Mode
    participant R as RealityKit Scene
    participant PS as Project Store
    participant E as Editor Module

    GA->>U: Show full doll with all placed assets selectable
    U->>GA: Taps an element
    GA->>U: Show transform controls (scale, rotation, position within bounds)
    U->>GA: Adjusts element
    GA->>R: Update transform (clamped to bounds)
    R-->>U: Updated 3D preview
    GA->>PS: Autosave
    alt Taps "Back to Slots"
        GA->>E: Return to slot editor (preserves current state)
    else Taps "Continue"
        GA->>U: Proceed to Scene Composer
    end
end
```

Steps:

1. **Global Adjustment → User** — Shows the fully assembled doll; each placed element is tappable
2. **User → Global Adjustment** — Taps an element to select it; transform controls appear
3. **User → Global Adjustment** — Adjusts scale, rotation, or position; all changes clamped to per-asset bounds
4. **Global Adjustment → RealityKit Scene** — Applies clamped transform updates in real time
5. **Global Adjustment → Project Store** — Autosaves after each adjustment
6. **User → Global Adjustment** — Taps "Back to Slots" to return to the slot-by-slot editor, or "Continue" to proceed to scene decoration

4. Scene Decoration Flow

mermaid

sequenceDiagram

```
participant U as User
participant SC as Scene Composer
participant C as Catalog Module
participant R as RealityKit Scene
participant PS as Project Store

SC->>C: Request background assets
C-->>SC: Background catalog
SC->>U: Show background catalog
U->>SC: Selects background
SC->>R: Apply background
SC->>U: Show foreground catalog
U->>SC: Selects foreground (or none)
SC->>R: Apply foreground
SC->>U: Show floating props catalog
U->>SC: Places props (or none)
SC->>R: Place props in scene
SC->>PS: Autosave
U->>SC: Taps "Continue"
```

Steps:

1. **Scene Composer → Catalog** — Loads background, foreground, and prop catalogs sequentially
2. **User → Scene Composer** — Selects background from catalog; preview updates in real time
3. **User → Scene Composer** — Selects foreground (or skips); selects and places floating decorative props (or skips)
4. **Scene Composer → Project Store** — Autosaves scene decoration state
5. **User → Scene Composer** — Taps "Continue" to proceed to music selection

5. Music Selection & Render Config Flow

mermaid

sequenceDiagram

participant U as User

participant MS as Music Selector

participant C as Catalog Module

participant PS as Project Store

MS->>C: Request music tracks

C-->>MS: Track catalog

MS->>U: Show track list + duration control (default 10s)

U->>MS: Selects track

MS->>U: Plays preview

U->>MS: Adjusts video duration

MS->>PS: Autosave music + duration choice

U->>MS: Taps "Render"

Steps:

1. **Music Selector → Catalog** — Loads available music-box tracks
2. **Music Selector → User** — Displays track list with play-preview and a video duration control (default 10 seconds)
3. **User → Music Selector** — Selects a track, listens to preview, adjusts duration
4. **Music Selector → Project Store** — Autosaves selection
5. **User → Music Selector** — Taps "Render" to trigger video generation

6. Render, Paywall & Share Flow

mermaid

sequenceDiagram

```
participant U as User
participant RE as Render Engine
participant R as RealityKit Scene
participant AV as AVFoundation
participant PW as Paywall Module
participant SK as StoreKit 2
participant SH as Share Flow
participant AN as Analytics

U->>RE: Trigger render
RE->>R: Capture 360° rotation frames
R-->>RE: Frame sequence
RE->>AV: Composite frames + audio → .mp4
AV-->>RE: Output video file URL
RE->>U: Show preview + "Share" button
RE->>PW: Check if first render
alt First render ever
    PW-->>RE: Allowed (free)
    RE->>U: Show note: "Future renders require purchase"
    U->>SH: Taps "Share"
    SH->>U: Open iOS share sheet
    AN->>AN: Log first_export event
else Subsequent render
    PW-->>RE: Blocked
    PW->>U: Show paywall (one-time purchase)
    alt User purchases
        U->>SK: Complete purchase
        SK-->>PW: Transaction confirmed
        PW->>U: Unlock export
        U->>SH: Taps "Share"
        SH->>U: Open iOS share sheet
        AN->>AN: Log purchase + export event
    else User cancels
        PW->>U: Return to preview (export blocked)
    end
end
end
```

Steps:

1. **User → Render Engine** — Triggers render from music selection screen
2. **Render Engine → RealityKit Scene** — Captures a 360° rotation over the configured duration
3. **Render Engine → AVFoundation** — Composites frame sequence with selected audio track into 1080×1920 .mp4

4. **Render Engine → User** — Shows video preview with "Share" button
 5. **Render Engine → Paywall Module** — Checks entitlement status
 6. **If first render:** Share is allowed immediately; user sees a note that future renders will require purchase; share sheet opens on tap
 7. **If subsequent render:** Paywall screen appears; user can complete one-time purchase via StoreKit 2 to unlock, or cancel and return to preview without exporting
 8. **Analytics** — Logs render, export, paywall impression, and purchase events
-

7. Draft Save & Resume Flow

mermaid

```
sequenceDiagram
    participant U as User
    participant App as Any Editor Screen
    participant PS as Project Store

    alt Autosave
        App->>PS: Save state on every meaningful change
    else Manual save/exit
        U->>App: Taps save/exit
        App->>PS: Save full project state + current step
        App->>U: Return to landing screen
    end

    Note over U,PS: On next launch, landing screen offers "Resume Draft"
```

Steps:

1. **App → Project Store** — Autosaves after every meaningful state change (asset selection, transform adjustment, scene/music choice)
 2. **User → App** — Can manually save and exit at any point; full state including current editor step is persisted
 3. **Landing Screen → User** — On next app launch, offers "Resume Draft" which loads saved state and navigates to the exact step the user left off
-

Acceptance Criteria

iOS UI / Navigation

- **AC-01** — App launches and landing animation begins playing within 2 seconds of tap on

iPhone 12 or newer (*REQ-01*)

- **AC-02** — Landing screen displays "Create New" button; if saved drafts exist, a "Resume Draft" option is also visible (*REQ-01, REQ-12*)
- **AC-03** — Resuming a draft restores the exact composition state (all slot selections, transforms, scene, music, duration) and navigates to the editor step where the user left off (*REQ-12*)
- **AC-04** — Maximum of 5 saved drafts; when the limit is reached, the user is prompted to delete an existing draft before creating a new one (*REQ-12*)
- **AC-05** — "Next" and "Back" buttons are functional on every slot screen; the user can navigate freely between all slots without losing selections (*REQ-04*)

Editor / 3D Composition

- **AC-06** — Each slot presents a scrollable catalog of available assets plus a "none" option; selecting "none" removes any previously placed asset for that slot (*REQ-02, REQ-03*)
- **AC-07** — Selecting an asset snaps it to the predefined anchor point with the default transform defined in the asset manifest (*REQ-05*)
- **AC-08** — Rotation controls clamp to the min/max rotation values from the asset manifest on all three axes; scale controls clamp to min/max scale values (*REQ-05*)
- **AC-09** — No freeform drag-positioning is available for core body parts; position offset is only available in global adjustment mode within bounded range (*REQ-05, REQ-06*)
- **AC-10** — Global adjustment mode displays all placed elements as tappable; selecting one shows transform controls; "Back to Slots" returns to the slot editor preserving all current state (*REQ-06*)
- **AC-11** — State autosaves after every asset selection or transform change; manually exiting saves full state including current step (*REQ-12*)

Scene & Music

- **AC-12** — Background, foreground, and floating prop catalogs each include a "none" option; scene preview updates in real time as selections change (*REQ-07*)
- **AC-13** — Music selection screen displays available tracks with playable previews and a duration slider; default duration is 10 seconds (*REQ-08*)
- **AC-14** — Adjusting the duration slider updates the stored video duration and is persisted with autosave (*REQ-08*)

Render & Export

- **AC-15** — Tapping "Render" produces a 1080×1920 vertical .mp4 video with the doll completing a 360° rotation over the configured duration, with the selected music track as

audio (*REQ-09*)

- **AC-16** — Render completes within 60 seconds on iPhone 12 (*REQ-09*)
- **AC-17** — After render, a video preview is displayed with a "Share" button (*REQ-09, REQ-10*)
- **AC-18** — Tapping "Share" opens the native iOS share sheet with the rendered .mp4 file attached; Instagram Reels/Stories appear as share destinations when Instagram is installed (*REQ-10*)

Paywall & Monetization

- **AC-19** — First render completes and share is accessible without any paywall; a visible note informs the user that future renders require purchase (*REQ-11*)
- **AC-20** — Second and all subsequent renders trigger the paywall screen before the share button is accessible (*REQ-11*)
- **AC-21** — Completing the one-time in-app purchase via StoreKit 2 immediately unlocks export; all future renders proceed without paywall (*REQ-11*)
- **AC-22** — Purchase state survives app deletion and reinstall; calling restore purchase recovers the entitlement (*REQ-15*)
- **AC-23** — If the user cancels the paywall, they return to the video preview screen; the rendered video is not lost but export remains blocked (*REQ-11*)

Content / Catalog

- **AC-24** — Phase 1 app ships with one doll template, 3–5 active slots, 20–30 assets, one scene template (background + foreground), and one music track bundled in the app package (*REQ-13*)
- **AC-25** — The entire creation, render, and share flow completes with no network connection (*REQ-13, REQ-14*)
- **AC-26** — No sign-in, account creation, or authentication prompt appears at any point in Phase 1 (*REQ-14*)

Analytics

- **AC-27** — TelemetryDeck events are logged for: app launch, editor start, each slot completion, global adjustment entry, scene decoration completion, music selection, render triggered, render completed, paywall shown, purchase completed, share initiated — all without collecting PII

Performance

- **AC-28** — App does not crash during a full creation-to-export flow on iPhone 12 with iOS 17

- **AC-29** — 3D preview in the editor maintains interactive frame rate (minimum 30 fps) during asset selection and transform manipulation on iPhone 12